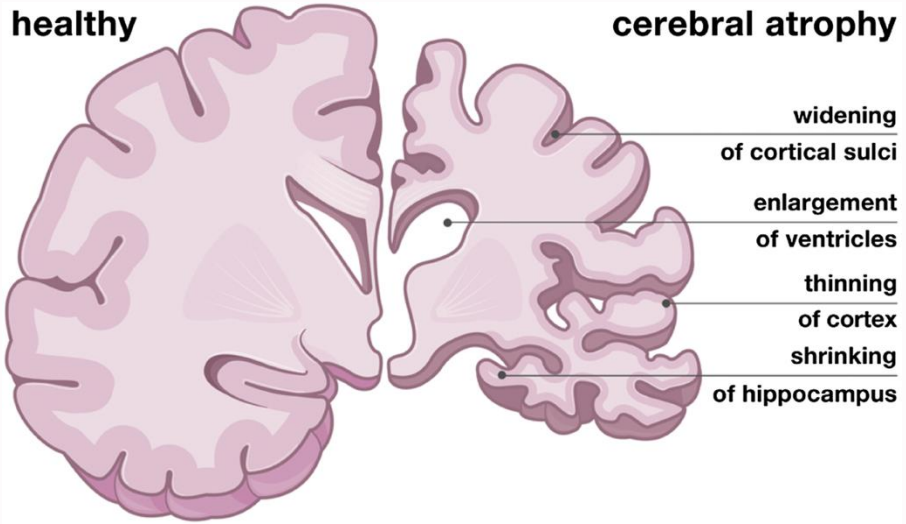
International Transfer of Stochastic Cortical Self-Reconstruction

Fabian Bongratz, Zhizheng Zhuo, Chao Zhang, Yaou Liu, Dennis M. Hedderich, and Christian Wachinger

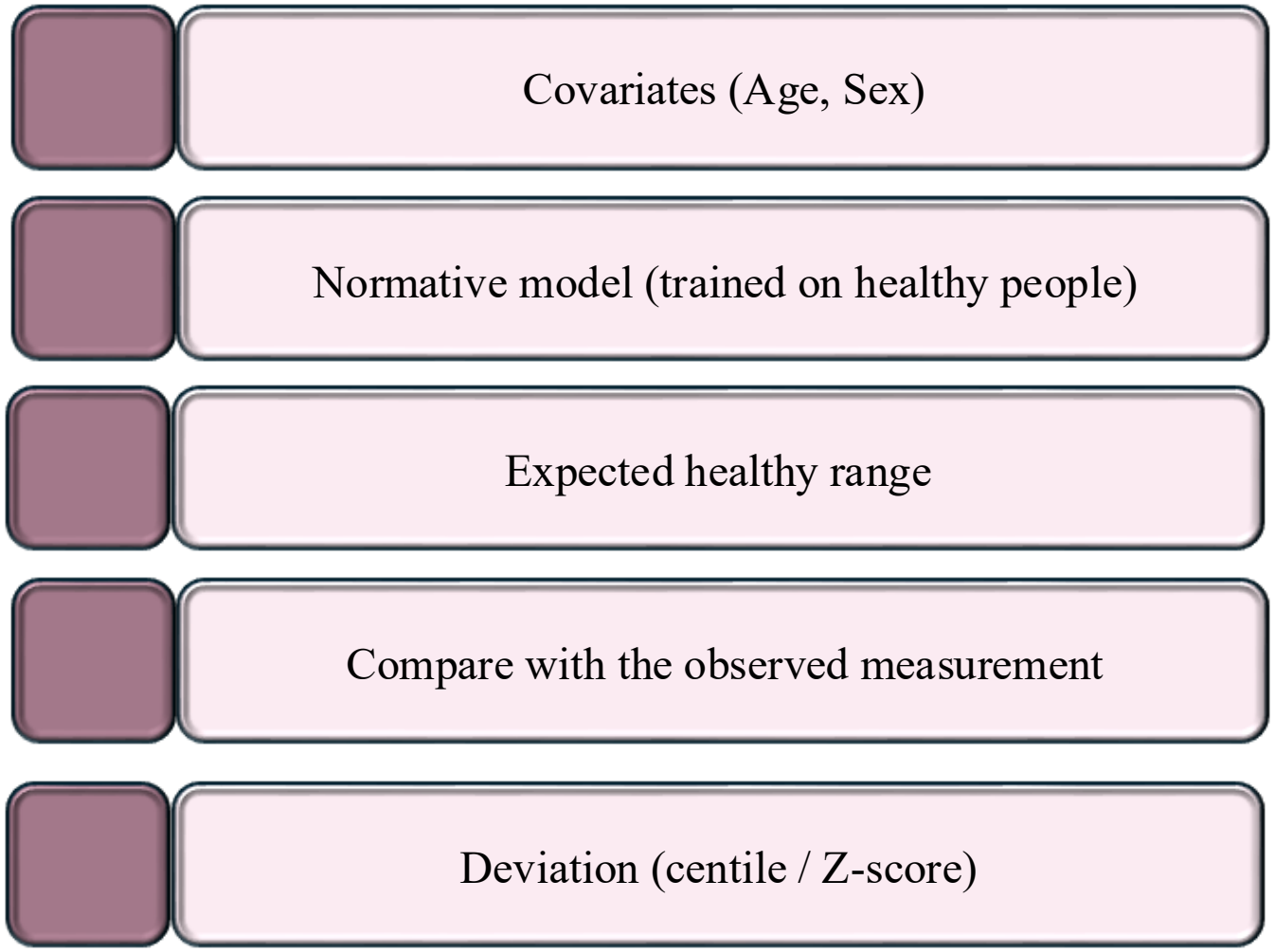
Aug 2026

Clinical Motivation: Why map cortical atrophy in each individual?

- **Gray matter atrophy** is a characteristic feature of neurodegenerative disorders such as Alzheimer's disease
- **Cortical thickness** is a sensitive biomarker for dementia and disease progression
- Atrophy patterns can be **subtle, spatially heterogeneous, and subject-specific**
- Individual-level assessment is challenging because of
  - **inter-subject anatomical variability**
  - **scanner/acquisition/software effects**



Conventional Normative Modeling: What is “normal” for this person?



Conventional Normative Modeling: Limitations

- Often based on **coarse regional measurements**
- Limited to the **covariates included during training**
- Reference is **not tailored to individual cortical morphology**
- Difficult to scale to **thousands of cortical vertices**

SCSR: Learning to Rebuild a Healthy Cortex

- Train only on healthy subjects
- Keep a random 20% of cortical vertices
- A neural network predicts the other 80%
- Repeat with different random subsets (stochastic)

SCSR: Personalized Reference and Z-score Map

- New subject: sample 20% of vertices and reconstruct the rest
- Repeat 100 times
- Reference = 95th centile of the reconstructions at each vertex
- $Z = (\text{observed} - \text{reference}) / \sigma$
- Lower Z = thinner than expected

Data: UK Biobank → Chinese Cohort

	UK Biobank	Chinese cohort
Role	Source (Original population)	Target (transfer)
Train / fine-tune	25,338	640
Validation	6,334	160
Age (years)	45-82	~4-85
Test	-	139 lifespan; 60 CN / 60 MCI / 60 AD

Preprocessing: MRI → FreeSurfer → FsAverage (left hemisphere) → ico5 mesh → 10,242 thickness values

Two Reconstruction Backbones in the Same SCSR Framework

Spherical UNet (SUNet)

Input: cortex as a surface (10,242 vertices)

Spherical convolutions on the cortical mesh

1.7M parameters

Neighbor relations built in

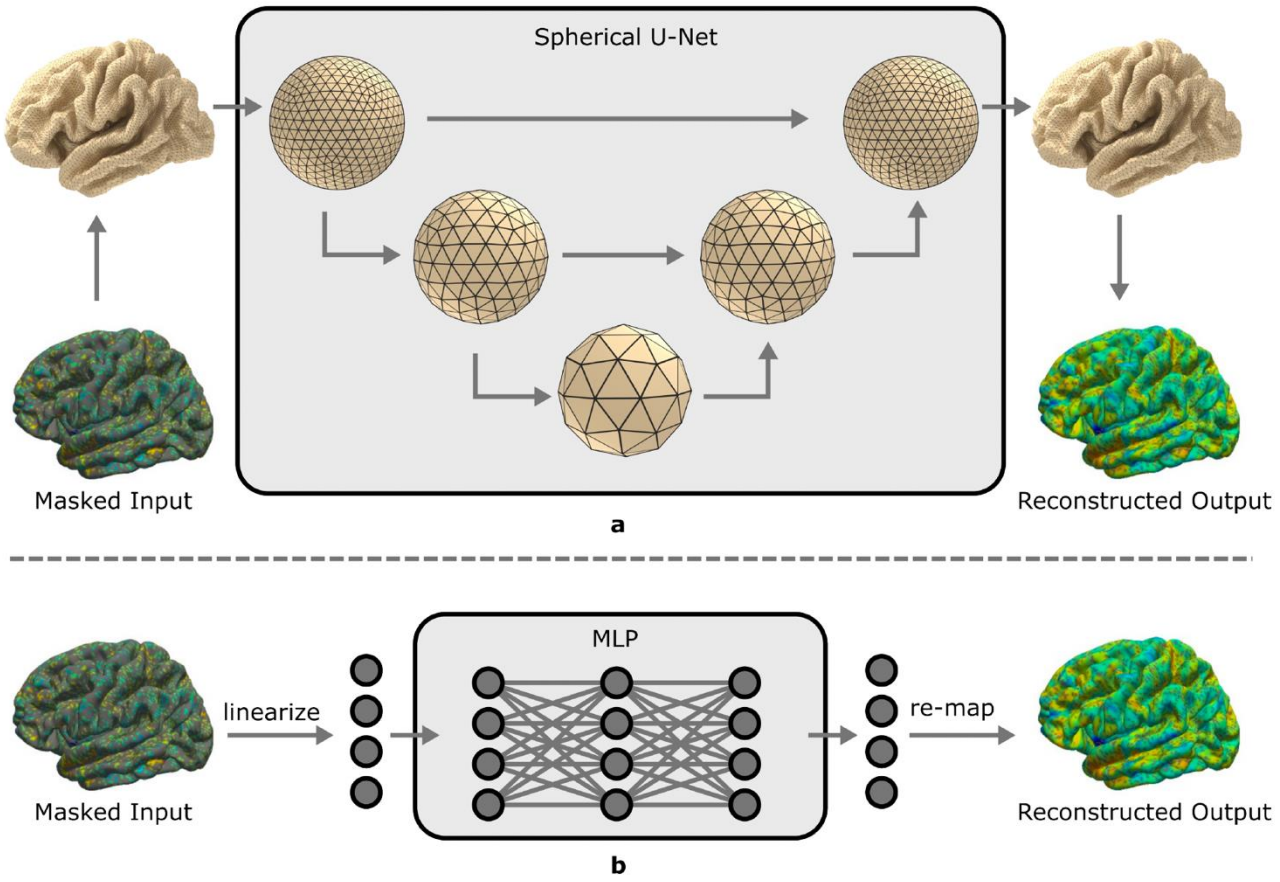
MLP

Input: cortex as a flat vector

Fully connected network

20M parameters

No spatial prior



SUNet: Zhao et al., 2019

How is SCSR transferred to the Chinese cohort?

- **Four adaptation strategies**

  Direct: UKB-trained model, no adaptation

  Fine-tuned: UKB model + Chinese training

  Scratch: trained only on Chinese data

  Joint: trained on pooled UKB + Chinese data

- **Applied to both MLP and SUNet → 8 configurations**

How are the models evaluated?

Reconstruction accuracy

MAE (mm) between observed thickness and SCSR reference

Healthy subjects only

By age group and overall; also on UKB validation

Lower is better

Atrophy detection

AUC of mean Z-score in AD-related ROI

AUC: how well the Z-score separates two groups (0.5 = chance, 1 = perfect)

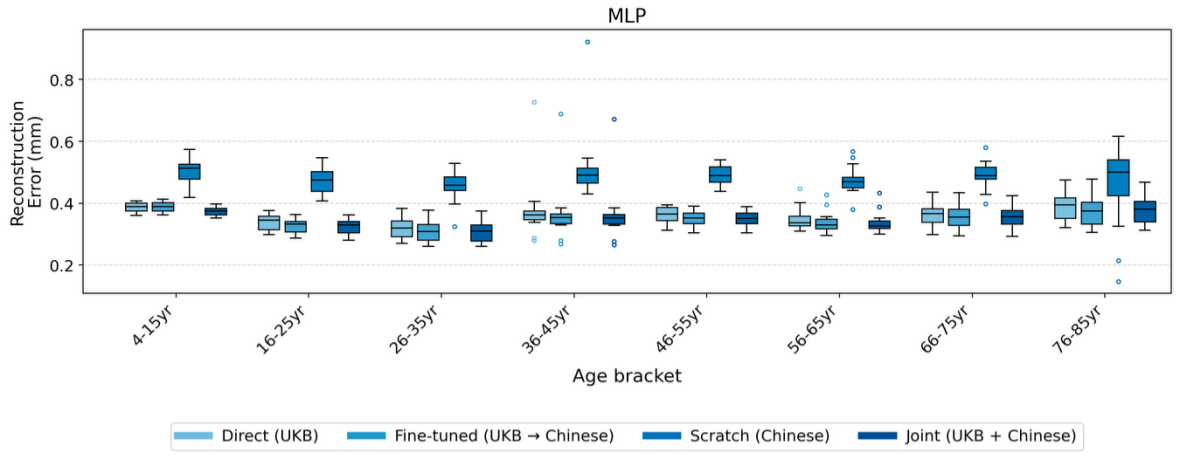
CN vs MCI · CN vs AD · MCI vs AD

Average of the three AUCs

Higher is better (0.5 = chance)

Reconstruction Performance Across Age

- Direct UKB ≈ adapted models (MLP: 0.36 vs 0.35 mm)
- Includes ages 4–20, outside UKB's range
- MLP-scratch overfits: 0.481 mm
- SUNet stable: 0.370–0.387 mm



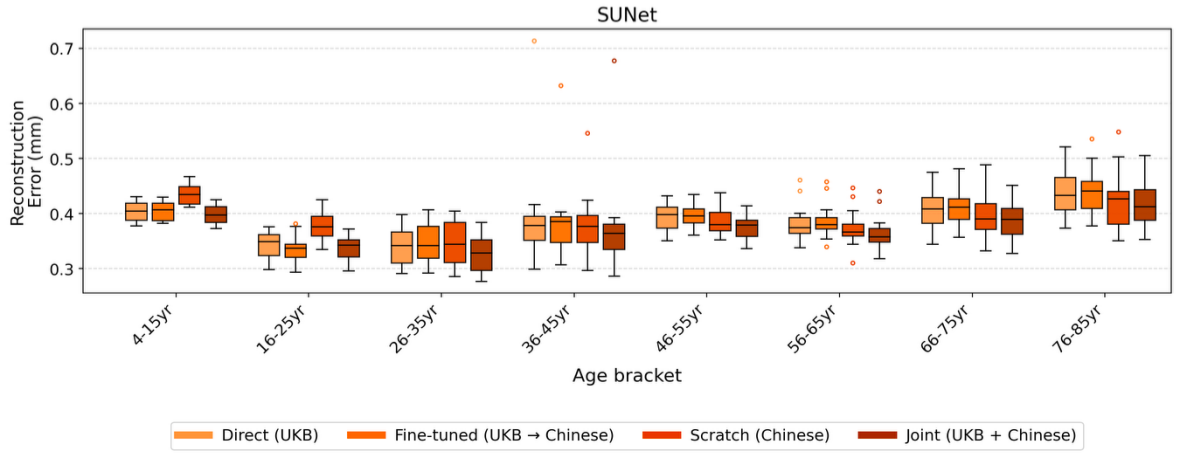


Table 1: Mean absolute reconstruction error (mm) on the Chinese lifespan test cohort (139 subjects) and the held-out UKB validation set (6,334 subjects), for all eight configurations. Best value per column in bold.

Backbone	Adaptation (Train Set)	Chinese	UKB
MLP	Direct (UKB)	0.359 ± 0.050	0.256 ± 0.025
	Fine-tuned (UKB → Chinese)	0.349 ± 0.048	0.285 ± 0.023
	Scratch (Chinese)	0.481 ± 0.070	0.454 ± 0.033
	Joint (UKB + Chinese)	**0.347 ± 0.047**	**0.250 ± 0.023**
SUNet	Direct (UKB)	0.385 ± 0.052	0.296 ± 0.020
	Fine-tuned (UKB → Chinese)	0.386 ± 0.050	0.329 ± 0.021
	Scratch (Chinese)	0.387 ± 0.046	0.348 ± 0.027
	Joint (UKB + Chinese)	0.370 ± 0.049	0.281 ± 0.020

Atrophy Detection: Can SCSR distinguish CN, MCI, and AD?

- SUNet > MLP in all four strategies
- Best: fine-tuned SUNet, average AUC 0.848
- Direct and joint SUNet are close (0.815, 0.817)
- CN vs MCI hardest; CN vs AD easiest

Table 2: Pairwise AUC for atrophy detection on the Chinese test cohort (60 CN, 60 MCI, 60 AD), based on mean SCSR Z-scores in the AD ROI. Best value per column in bold.

Backbone	Adaptation (Train Set)	CN|MCI	CN|AD	MCI|AD	Average
MLP	Direct (UKB)	0.543	0.841	0.775	0.720
	Fine-tuned (UKB → Chinese)	0.581	0.849	0.765	0.732
	Scratch (Chinese)	0.598	0.789	0.711	0.699
	Joint (UKB + Chinese)	0.524	0.801	0.753	0.693
SUNet	Direct (UKB)	0.718	0.925	0.803	0.815
	Fine-tuned (UKB → Chinese)	**0.787**	**0.938**	**0.818**	**0.848**
	Scratch (Chinese)	0.631	0.863	0.801	0.765
	Joint (UKB + Chinese)	0.698	0.928	0.826	0.817

Spatial Patterns of Cortical Atrophy

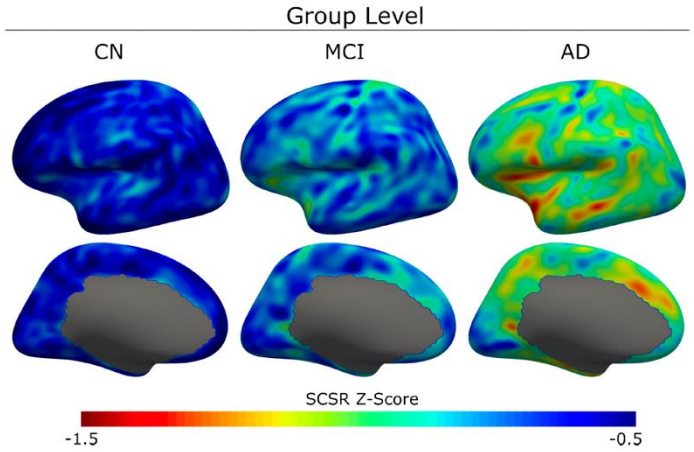


Fig. 3: Vertex-wise group mean SCSR Z-scores for the three diagnostic groups: age-matched cognitively normal (CN), mild cognitive impairment (MCI), and Alzheimer's disease (AD). The visualization was created based on the finetuned SUNet model. Lower Z-scores indicate more severe atrophy. The medial region that connects the two brain hemispheres is not included in SCSR and is therefore grayed out in the visualization. Rendering was performed using PyVista.

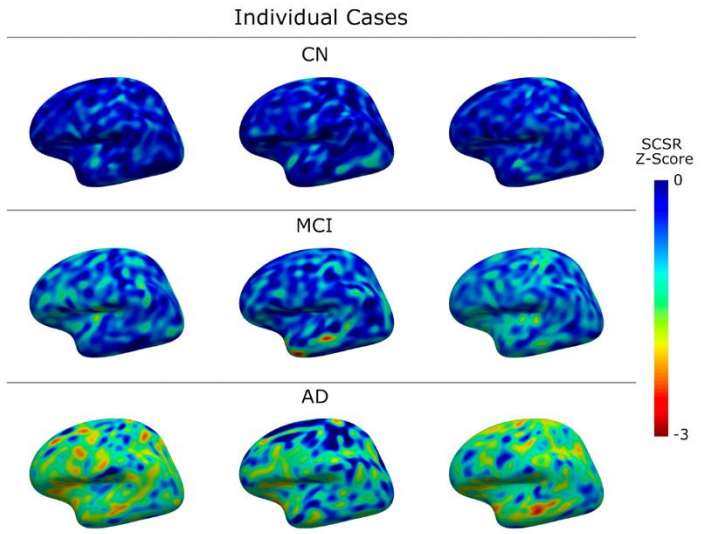


Fig. 4: Individual SCSR Z-score maps computed based on three subjects from each diagnostic group (CN, MCI, AD), respectively. The visualization was created based on the finetuned SUNet model. Lower Z-scores indicate more severe atrophy. Rendering was performed using PyVista.

- AD: strongest thinning in temporoparietal, medial temporal, frontal regions
- MCI: milder, similar pattern
- Lower Z = thinner than the healthy reference

- Large differences in extent and severity within each group

Conclusions

- UKB-trained SCSR transfers to a Chinese cohort without adaptation, even for ages 4–20
- SUNet detects atrophy better than MLP and is more robust with limited data
- Fine-tuning: best detection (AUC 0.848). Joint training: best reconstruction
- Without target data: apply the UKB model directly

Thank you ☺

Kaveh Moradkhani